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Liquid containers, filters, caps, hose assemblies and kits

Abstract

A container may include a generally hollow body configured for holding a liquid and including a spout. A spout cap may be configured for removable engagement with the spout. The spout cap may include an inner boss and an outer boss in fluid communication with the inner boss. The inner boss may include an annular skirt disposed within an interior space of the spout cap. The outer boss may be configured to receive a connector of a hose assembly. A filter may have a connecting portion configured for removable insertion into the inner boss. The filter and the spout cap may form a fluid flow path from the body through the spout. Related filters, hose assemblies, and kits are also described.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
596369	12/1896	Jackson	210/466	A47J 31/14
2556627	12/1950	Miksis	N/A	N/A
2723056	12/1954	Smith	222/401	B65D 47/061
4583668	12/1985	Maynard, Jr.	215/390	B65D 25/48
D301689	12/1988	Fick et al.	N/A	N/A
D305982	12/1989	Gillispie et al.	N/A	N/A
5076333	12/1990	Law	141/335	B65D 47/283
D338158	12/1992	Poirier	N/A	N/A
D340640	12/1992	Ferguson et al.	N/A	N/A
5249611	12/1992	Law	141/335	B67D 3/04
D346113	12/1993	Ferguson et al.	N/A	N/A
D358330	12/1994	Kahl	N/A	N/A
5472025	12/1994	Conrad et al.	N/A	N/A
5609759	12/1996	Nohren, Jr	210/474	C02F 1/003
6006961	12/1998	Wark	N/A	N/A
D433635	12/1999	Chrisco et al.	N/A	N/A
D434328	12/1999	Chrisco et al.	N/A	N/A
D477224	12/2002	Hussain	N/A	N/A
D482967	12/2002	Bourque et al.	N/A	N/A
D516434	12/2005	McGuire	N/A	N/A
D526203	12/2005	Mishra	N/A	N/A
D528423	12/2005	Glover	N/A	N/A
D534796	12/2006	Falkenburg	N/A	N/A
D542655	12/2006	Szczesniak	N/A	N/A
7240809	12/2006	Godfrey	N/A	N/A
7416087	12/2007	Press	N/A	N/A
D581481	12/2007	Edwards	N/A	N/A
D636056	12/2010	Pritchard	N/A	N/A
D654796	12/2011	Wurth et al.	N/A	N/A
D685452	12/2012	Turner	N/A	N/A
D732393	12/2014	Senus	N/A	N/A
D743264	12/2014	Mariani	N/A	N/A

D760345	12/2015	Campbell et al.	N/A	N/A
D760869	12/2015	Sanders	N/A	N/A
D763696	12/2015	Eisen	N/A	N/A
D782321	12/2016	Ihorovych	N/A	N/A
D815943	12/2017	Candelaria	N/A	N/A
D866338	12/2018	Eisen	N/A	N/A
D919436	12/2020	Currie et al.	N/A	N/A
D935322	12/2020	Darbyshire et al.	N/A	N/A
D941147	12/2021	Wang	N/A	N/A
D942273	12/2021	Bravman et al.	N/A	N/A
D984641	12/2022	Fuller	N/A	N/A
2006/0081652	12/2005	Press	222/189.06	B67D 7/005
2010/0078094	12/2009	Trippi, Jr.	141/352	B67D 7/005
2012/0325858	12/2011	MacDonald	222/189.07	B65D 25/48
2019/0106314	12/2018	Shaw	N/A	B65D 85/84
2022/0194663	12/2021	Taylor et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	**Application Date**	**Country**	**CPC**
302241160	12/2011	CN	N/A
307000366	12/2020	CN	N/A
256185-001	12/2013	IN	N/A

OTHER PUBLICATIONS

Polisport 20L Pro Octane Rapid Fuel Can, https://www.wscperformance.co.uk/products/polisport-20l-pro-octane-rapid-fuel-can, webpage accessed on Feb. 22, 2023 (2 pages). cited by applicant
Polisport 20L Pro Octane Rapid Fuel Can, http://web.archive.org/web/20220705085001/https://www.wscperformance.co.uk/products/polisport-201-pro-octane-rapid-fuel-can as of Jul. 5, 2022, Wayback Machine webpage accessed Feb. 22, 2023 (2 pages). cited by applicant
Hyuduo, Aug. 6, 2020, Amazon.com Product Post, URL: https:// www.amazon.com/Strainer-Pressure-Accessories-Light-Weight-Accessory/dp/B08FCHVQPG (year :2020). cited by applicant

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 18/195,280 filed May 9, 2023, the disclosure of which is incorporated herein by reference.

FIELD
(1) This application relates to liquid containers, such as liquid fuel containers.
BACKGROUND
(2) In the field of liquid fuel containers, it is often desirable to filter the liquid fuel before it is poured into a fuel tank of a vehicle, for example, to prevent unwanted particulates from fouling the fuel system, engine, or other components of the vehicle. However, there is often no convenient and easy way to filter the liquid fuel before it is poured into the fuel tank, particularly in situations such

as races wherein time is critical. Also, if a filter is available, it is prone to becoming clogged with particulates. Additionally, although some containers offer some type of filtration, not only is the filter not replaceable, but the screen size of the filter is typically so large that it does not actually filter out the critical, smaller particulates that could be harmful to motors. And because there is no practical way to swap out the filter, it is impossible to switch to a smaller (e.g., lower micron) screen size that would be effective. It would be a significant advancement in the art to provide a simple, quick, and easy way to filter a liquid fuel or other liquid before it is poured into a fuel tank or other liquid repository. It would also be a significant advancement to provide a filter that is easily replaceable if it becomes clogged with particulates or if a different screen size is needed.

SUMMARY

(3) In some embodiments, a container may include a generally hollow body configured for holding a liquid and including a spout. A spout cap may be configured for removable engagement with the spout. The spout cap may include an inner boss and an outer boss in fluid communication with the inner boss. The inner boss may include an annular skirt disposed within an interior space of the spout cap. The outer boss may be configured to receive a connector of a hose assembly. A filter may have a connecting portion configured for removable insertion into the inner boss. The filter and the spout cap may form a fluid flow path from the body through the spout.

(4) In some embodiments, a filter may include a connecting portion having a conduit extending therethrough and configured for removable insertion into a receptacle, and at least one mesh screen in fluid communication with the conduit, wherein the at least one mesh screen and the conduit form a fluid flow path through the filter. In some embodiments, the filter may include a base; a head; a plurality of columns connecting the base and the head; at least one mesh screen disposed in gaps between the plurality of columns, the base, and the head; a connecting portion extending from the head, the connecting portion configured for removable insertion into a receptacle; and a conduit extending through the head and the connecting portion; wherein the at least one mesh screen and the conduit form a fluid flow path through the filter. In some embodiments, the at least one mesh screen may include a substantially flat mesh screen disposed substantially perpendicular to a central axis of the conduit. In some embodiments, the filter may include a head and a plurality of arches extending from the head and joined together at a central position above the head, wherein the at least one mesh screen is disposed in gaps between the plurality of arches. In some embodiments, the filter may include a rim adjacent to the connecting portion, a base spaced apart from the rim, and a plurality of columns connecting the base and the rim, wherein the at least one mesh screen is disposed in gaps between the plurality of columns, the base, and the rim, and wherein the conduit extends through the rim and the connecting portion. In some embodiments, the filter may have a screen with a conical shape, which may or may not have one or more support elements attached thereto.

(5) In some embodiments, a kit may include a container having a generally hollow body configured for holding a liquid and including a spout and a vent. The kit may include a spout cap configured for removable engagement with the spout, the spout cap including an inner boss and an outer boss in fluid communication with the inner boss, the inner boss being disposed within an interior space of the spout cap, the outer boss configured to receive a connector of a hose assembly. The kit may include a vent cap removably engaged with the vent. The kit may include one or more filters each having a connecting portion configured for removable insertion into the inner boss, and one or more hose assemblies each having a connector configured for removable engagement with the outer boss. When a selected filter from the one or more filters is installed in the inner boss and a selected hose assembly from the one or more hose assemblies is installed in the outer boss, the selected filter, the spout cap, and the selected hose assembly form a fluid flow path from the body through the selected filter, the spout cap, and the selected hose assembly.

(6) In some embodiments, a container may include a generally hollow body configured for holding a liquid and including a spout, a vent, and a handle. The handle may have a front end and a rear

end. The spout may be disposed in front of the handle, and the vent may be disposed at or proximate to the rear end of the handle. In some embodiments, the vent may be substantially aligned with a central axis of the handle. In some embodiments, the vent may face rearward from the handle. In some embodiments, a vent cap may be removably engaged with the vent. In some embodiments, the handle and the vent cap may be configured for one-handed grasping of the handle and manipulation of the vent cap.
(7) In some embodiments, a spout cap is provided for removable engagement with a spout of a container. The spout cap may include an upper portion; an outer skirt depending downward from the upper portion and including internal threads configured for threaded engagement with external threads on the spout; an inner boss depending downward from the upper portion and including an annular skirt disposed within an interior space of the spout cap, the inner boss configured for receiving a connecting portion of a filter; and an outer boss depending upward from the upper portion and configured to receive a connector of a hose assembly, the outer boss being in fluid communication with the inner boss.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a perspective view of a liquid fuel container with a spout cap and a vent cap shown in exploded positions.
(2) FIG. **2** is a cross-sectional view of the spout cap of FIG. **1**.
(3) FIG. **3** is an exploded perspective view of the spout cap of FIG. **1** and a fuel filter.
(4) FIG. **4** is an exploded perspective view of the spout cap of FIG. **1** and a hose assembly.
(5) FIG. **5** is a cross-sectional view of the spout cap and fuel filter of FIG. **3** showing the fuel filter in an installed position and showing a spout plug in an exploded position. The mesh screen of the fuel filter is not shown in FIG. **5** for clarity.
(6) FIG. **6** is a side elevational view of the fuel container, spout cap, and vent cap of FIG. **1** with the hose assembly of FIG. **4** coupled to the spout cap and illustrating manual operation of the vent cap.
(7) FIG. **7** is a cross-sectional view of another spout cap and fuel filter.
(8) FIG. **8** is a perspective view of another filter.
(9) FIG. **9** is a cross-sectional view of the filter of FIG. **8**.
(10) FIG. **10** is a perspective view of yet another filter.
(11) FIG. **11** is a cross-sectional view of the filter of FIG. **10**.
(12) FIG. **12** is a perspective view of yet another filter.
(13) FIG. **13** is a cross-sectional view of the filter of FIG. **12**.
(14) FIG. **14** is a perspective view of yet another filter.
(15) FIG. **15** is a cross-sectional view of the filter of FIG. **14**.
DETAILED DESCRIPTION
(16) The following terms as used herein should be understood to have the indicated meanings:
(17) When an item is introduced by “a” or “an,” it should be understood to mean one or more of that item.
(18) “Comprises” means includes but is not limited to.
(19) “Comprising” means including but not limited to.
(20) “Having” means including but not limited to.
(21) As shown in FIG. **1**, a liquid fuel container **10** may have a generally hollow body **12** adapted for holding liquid fuel. Container **10** may include a spout **14** and a spout cap **16** configured for removable engagement with the spout. For example, spout **14** may have external threads configured for threaded engagement with internal threads of spout cap **16**. Container **10** may also have a vent **18** and a vent cap **20** configured for removable engagement with the vent. For example,

vent **18** may have external threads configured for threaded engagement with internal threads of vent cap **20**. To facilitate handling and use, container **10** may include a top handle **22**, a rear handle **24** (see FIG. **6**), and a finger hold indentation **26**. Of course, other handle or grip arrangements may be used, if desired. As shown, spout **14** may be disposed in front of handle **22**, and vent **18** may be disposed at or proximate to the rear end of handle **22**. In some embodiments, vent **18** may be substantially aligned with a central axis of handle **22**, and vent **18** may face rearward from handle **22**. In some embodiments, the walls of container **10** may include one or more recesses **28** that may serve as wall stiffeners as well as gripping aids. Similarly, some embodiments may include one or more ridges **30** along one or more edges of container **10**, which again may help stiffen the walls and aid in gripping the container.

(22) Referring to FIG. **2**, spout cap **16** may have an outer skirt **65** extending downward from an upper portion **67**, the outer skirt **65** including internal threads **58** configured for threaded engagement with external threads **69** on spout **14** (see FIG. **1**). Spout cap **16** may have an outer boss **50** and an inner boss **54** configured with threads **52** and **56**, respectively. Threads **52** and **56** may be separated by an unthreaded portion **64** that may or may not serve as a stop. In some embodiments, threads **52** and **56** may be opposing threads oriented in opposite thread directions (e.g., threads **52** may be right-handed or left-handed from the exterior, and threads **56** may be right-handed or left-handed from the interior), or threads **52** and **56** may be oriented in the same thread direction. As shown, outer boss **50** may depend upward from upper portion **67**, and inner boss **54** may be in the form of an annular skirt projecting downward from upper portion **67** such that inner boss **54** is disposed within an interior space of spout cap **16**. Threads **52** may be configured for fluid-tight threaded engagement with threads **33** of a connector **34** of hose assembly **32** as described further below. Threads **56** may be configured for fluid-tight threaded engagement with threads **82** of a filter **70** as described further below. Spout cap **16** may also include an O-ring **62** or flat gasket seated in an annular groove **60** in upper portion **67** for fluid-tight engagement with the rim of spout **14**. Of course, any suitable fluid-tight sealing arrangement may be used. In some embodiments, as shown in FIG. **5**, a plug **42** having a suitable head (e.g., hex head **44**) and threads **46** may be screwed into the threads **52** of outer boss **50** to close spout **14** when the hose assembly **32** is not in use. The threads described herein may be configured for fluid-tight sealing engagement of the various components. For example, in some embodiments, the threads may be NPT standard pipe threads or other suitable threads.

(23) As shown in FIG. **3**, filter **70** may have a base **72** and a plurality of columns **74** extending between base **72** and a hex head **80**. A mesh screen **76** (or a plurality of mesh screens) may be disposed in the gaps between the columns **74** to filter out particulates from the liquid as it is being poured from container **10** to prevent the particulates from entering the repository into which the liquid is poured (e.g., a fuel tank of a vehicle). For example, a suitable mesh screen **76** may be inter-molded with or attached to the columns **74**, hex head **80**, and/or base **72**. The mesh size of mesh screen **76** may be selected to filter a desired size range of particulates and to achieve a desired flow rate when liquid is poured from container **10**. Hex head **80** may be used to manually grasp filter **70** and screw it into inner boss **54**. As shown in FIG. **5**, filter **70** thus forms part of a fluid flow path **84** from the interior of container **10** through mesh screen **76** and conduit **78** of filter **70**, and the fluid continues outward through outer boss **50** of spout cap **16** and through the hose assembly **32**. Filter **70** is thus protected inside container **10** and stays clean and out of the way, and no extra housing is needed to protect the filter.

(24) As shown in FIG. **4**, the hose assembly **32** may include a hose connector **34** configured for threaded, fluid-tight engagement (e.g., via NPT standard pipe threads or other suitable threads **33**) with threads **52** of outer boss **50** of spout cap **16**. A hose **36** may extend from hose connector **34** to a nozzle **38**. In some embodiments, nozzle **38** may have a threaded end fitting **35** (see FIG. **6**) configured for accepting a threaded hose cap **40** when hose assembly **32** is not in use. In some embodiments, hose cap **40** and vent cap **20** may be interchangeable. Of course, other suitable hose

assemblies may be used.

(25) Another embodiment of a spout cap **116** and filter **170** is shown in FIG. **7**. In this embodiment, spout cap **116** is similar to spout cap **16** described above except that inner boss **54** has a smooth cylindrical internal bore with a groove **108** rather than threads **56**. Filter **170** is similar to filter **70** described above except that, rather than threads **82** on the connecting portion of filter **70**, the connecting portion of filter **170** has a smooth cylindrical shank **106** with a groove **104** in which an O-ring **102** is disposed. O-ring **102** is configured to snap into groove **108** and provide a liquid-tight seal between filter **170** and spout cap **116** when filter **170** is pressed into inner boss **54**. Filter **170** may be pulled and removed from inner boss **54** for cleaning or replacement. Of course, other sealing arrangements for the filter and spout cap may be employed if desired.

(26) As illustrated in FIG. **6**, with hose assembly **32** installed in outer boss **50** of spout cap **16** as shown and the hose cap **40** removed, nozzle **38** of hose assembly **32** may be inserted into an opening of a desired repository (e.g., a fuel tank of a vehicle), and container **10** may be tilted over from its upright position to pour liquid from container **10** into the repository. In some embodiments, hose assembly **32** may include a stop, such as tapered stop **45** shown in FIG. **4**, configured to engage with the opening of the repository and help to properly position nozzle **38** in the opening. The stop **45** may also transfer some of the weight of the liquid-filled container **10** away from the user, allowing it to rest slightly on the repository. In some embodiments, a user may grasp container **10** in a “backwards” fashion as shown in FIG. **6** so that the user may twistingly operate the vent cap **20** (as shown at **21**) with the thumb and forefinger of the user's same hand to control the amount of venting and thus the rate of fluid flow from the container. The user may grasp the rear handle **24** or indentation **26** with the user's other hand for stability and control of the container during pouring operations. The configuration of the vent **18** and vent cap **20** as shown may mitigate against any awkward jostling of the container **10** as a user seeks to open the vent **18** once the hose **36** is inserted into the repository. With other types of containers, a user generally must awkwardly hold the container with one hand at its center of gravity in order to free up the user's other hand to open the vent. With the present container, a user may readily open the vent while firmly keeping a grasp on opposing ends of the container, which results in safer, steadier, and more controlled use of the container.

(27) Referring to FIGS. **8-15**, other alternative filters are shown having threads **82** configured for threaded engagement with threads **56** of spout cap **16** similar to filter **70** described above. For example, as shown in FIGS. **8-9**, filter **270** may have a hex head **80** and a plurality of arches **272** extending from hex head **80** and joined together at a central position above hex head **80**. Similar to filter **70**, a mesh screen **76** or a plurality of mesh screens (not shown in FIG. **9** for clarity) may be disposed in the gaps between the arches **272** to filter out particulates from the liquid as it is being poured from container **10** through conduit **78** to prevent the particulates from entering the repository into which the liquid is poured (e.g., a fuel tank of a vehicle). For example, a suitable mesh screen **76** may be inter-molded with or attached to the arches **272** and/or hex head **80**. The mesh size of mesh screen **76** may be selected to filter a desired size range of particulates and to achieve a desired flow rate when liquid is poured from container **10**. Hex head **80** may be used to manually grasp filter **270** and screw it into inner boss **54** of spout cap **16** similar to filter **70** described above.

(28) As illustrated in FIGS. **10-11**, instead of a hex head, filter **370** may have a multi-lobed head **380** for manual grasping of filter **370** to screw it into inner boss **54** of spout cap **16** via threads **82**. Similar to filter **70**, filter **370** has a plurality of columns **74** extending between multi-lobed head **380** and a base **72**, and a mesh screen **76** or a plurality of mesh screens (not shown in FIG. **11** for clarity) may be disposed in the gaps between the columns **74** to filter out particulates from the liquid. Filter **370** has an internal conduit **78** through which liquid may flow similar to filter **70** described above. Also similar to filter **70**, a suitable mesh screen **76** may be inter-molded with or attached to the columns **74**, base **72**, and/or multi-lobed head **380**. The mesh size of mesh screen **76**

may be selected to filter a desired size range of particulates and to achieve a desired flow rate when liquid is poured from container **10**.

(29) Referring to FIGS. **12-13**, another alternate filter **470** may have a multi-lobed head **380** similar to filter **370** described above, and a substantially flat mesh screen **76** may be disposed across the opening of conduit **78** (e.g., substantially perpendicular to the central axis of conduit **78**) on or adjacent to the multi-lobed head **380**. As with other embodiments described herein, the mesh size of mesh screen **76** may be selected to filter a desired size range of particulates and to achieve a desired flow rate when liquid is poured from container **10** through filter **470**. The mesh screen **76** may be inter-molded with or attached to multi-lobed head **380** or another portion of filter **470** adjacent thereto. In some embodiments, filter **470** may have a hex head rather than a multi-lobed head, or a combination thereof.

(30) Referring to FIGS. **14-15**, another alternative filter **570** may have a rim **582** adjacent threads **82**, a base **72** spaced from rim **582**, and a plurality of columns **74** extending between rim **582** and base **72**. A hex head **580** may extend from base **72** to facilitate manual grasping of filter **570**. Filter **570** has an internal conduit **78** through which liquid may flow similar to other embodiments described above. Also similar to other embodiments, a mesh screen **76** or a plurality of mesh screens (not shown in FIG. **15** for clarity) may be disposed in the gaps between the columns **74** to filter out particulates from the liquid. Mesh screen **76** may be inter-molded with or attached to the columns **74**, base **72**, and/or rim **582**. The mesh size of mesh screen **76** may be selected to filter a desired size range of particulates and to achieve a desired flow rate when liquid is poured from container **10**.

(31) Alternatively, instead of threads **82**, each of filters **270**, **370**, **470**, **570** may be provided with a smooth cylindrical shank **106** with a groove **104** in which an O-ring **102** is disposed as described above for filter **170** and thus configured for a press-fit, removable engagement with inner boss **54** of spout cap **116** (see FIG. **7**). In such embodiments, O-ring **102** is configured to snap into groove **108** and provide a liquid-tight seal between filter **270**, **370**, **470**, **570** and spout cap **116**, and filter **270**, **370**, **470**, **570** may be pulled and removed from inner boss **54** for cleaning or replacement.

(32) In some embodiments, additional advantages may be realized if the threads of filter **70** and connector **34** of hose assembly **32** are the same or compatible. For example, if filter **70** is not installed in inner boss **54**, the hose assembly **32** (via connector **34**) may be screwed into inner boss **54** before spout cap **16** is screwed onto spout **14**, and the hose assembly **32** may be suitably sized and disposed inside container **10**, such as during initial shipment or storage, for example. If desired, the filter **70** may also be disposed inside container **10** (e.g., in a loose, uninstalled condition) and then installed in inner boss **54** when ready to use. Alternatively, filter **70** may be installed in spout cap **16**, and hose assembly **32** may be disposed inside container **10** in a loose, uninstalled condition; or, both filter **70** and hose assembly **32** may be disposed inside container **10** in a loose, uninstalled condition awaiting installation and use as described herein.

(33) In some embodiments, a container **10** may be provided as a kit (e.g., a fueling kit) that includes one or more hose assemblies **32**, one or more filters **70**, and one or more plugs **42** in addition to container **10**, spout cap **16**, and vent cap **20**. For example, a kit may include a plurality of hose assemblies **32** having different sizes and flow rates and a plurality of filters **70** having different mesh sizes and flow rates. A user may select a particular hose assembly **32** and a particular filter **70** from among the respective pluralities of hose assemblies and filters to achieve a desired flow rate and degree of filtration, depending on the user's needs.

(34) Although the containers and other components illustrated herein have been described primarily for use as fuel containers and fueling assemblies, persons of ordinary skill in the art will appreciate that such containers and components may also be used for other liquids, such as oil, water, solvents, chemicals, beverages, and other liquid compositions. The containers and other components described herein may be made of suitable materials, depending on the particular liquids involved. For example, in some embodiments designed for liquid fuels, the containers and

other components illustrated herein may be made of suitable plastic, such as high density polyethylene (HDPE). The hose **36** may be any suitable hose material, such as PVC with dibutyl and dioctyl phthalate plasticizers, polyurethane, fluorosilicone, or PVDF, for example. Although threaded connections have been described for the various connections between and among the filters, spout caps, containers, spouts, and hose assemblies described herein, other suitable connections, such as quick connect fittings, for example, may also be used to connect those components.

(35) Although the structures disclosed herein and some of their advantages have been described in detail, it should be understood that various changes, substitutions, and alterations may be made herein without departing from the invention as defined by the appended claims and their legal equivalents. For example, among other things, any feature described for one embodiment may be used in any other embodiment, and any feature described herein may be used independently or in combination with other features. Moreover, the scope of the present application is not intended to be limited to the particular embodiments described in the specification. Use of the word “include,” for example, should be interpreted as the word “comprising” would be, i.e., as open-ended. As one will readily appreciate from the disclosure, processes, machines, manufactures, compositions of matter, means, methods, or steps presently existing or later to be developed that perform substantially the same function or achieve substantially the same result as the corresponding embodiments described herein may be utilized. Accordingly, the appended claims are intended to include within their scope such processes, machines, manufactures, compositions of matter, means, methods or steps.

Claims

1. A spout cap for removable engagement with a spout of a container, the spout cap comprising: an upper portion; an outer skirt depending downward from the upper portion and including internal threads configured for threaded engagement with external threads on the spout; an inner boss depending downward from the upper portion and including an annular skirt disposed within an interior space of the spout cap, the inner boss configured for receiving a connecting portion of a filter; and an outer boss depending upward from the upper portion and configured to receive a connector of a hose assembly, the outer boss being in fluid communication with the inner boss.
2. The spout cap of claim 1 wherein the upper portion includes an annular groove, and further comprising an O-ring or flat gasket disposed in the groove.
3. The spout cap of claim 1 further including a filter removably engaged in the inner boss.
4. The spout cap of claim 3 wherein the filter and the inner boss are configured for threaded engagement.
5. The spout cap of claim 3 wherein: the filter includes a connecting portion having a smooth cylindrical shank with a groove in which an O-ring is disposed; and the inner boss includes an internal groove in which the O-ring is also disposed.
6. The spout cap of claim 1 wherein the inner boss and the outer boss are configured with opposing threads adapted for threaded engagement with the connecting portion of the filter and the connector of the hose assembly, respectively.
7. The spout cap of claim 6 wherein the opposing threads are NPT threads.